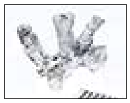
New Zealand lizard's extinction
The Oligosoma infrapunctatum, a New Zealand lizard is now extinct just after 130 years of existence. **https://dgit.in/sep19-11**



Printing living tissue in seconds
Researchers came up with an optical method to make complex live tissue utilising bioprinting. **https://dgit.in/sep19-12**

top, forming what is now the crust of the Moon. During this cooling period, various kinds of rocks were formed, some of which are still on the surface of the Moon today. Now the question is, exactly when this collision event took place.

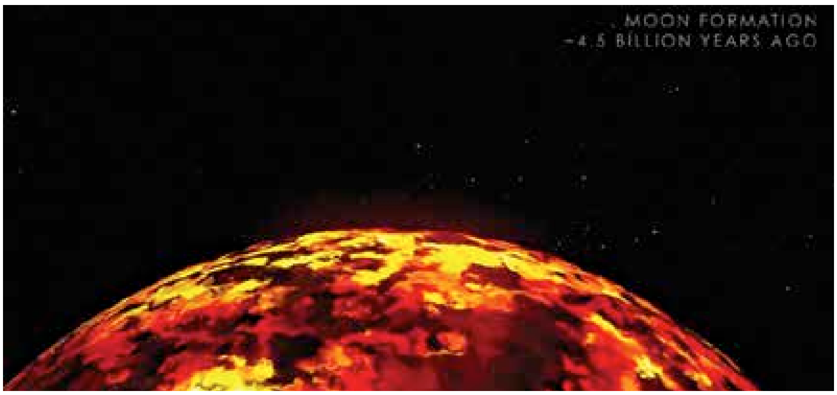
There were two proposed time windows during which the impact and the subsequent cooling of the Moon apparently took place. One was about a 100 million years while the other was about 200 million years after the formation of the Solar System.

In 2017, scientists from the Solar System Exploration Research Virtual Institute (SSERVI) published a study that indicated that the Moon had formed earlier than previously thought. While there were several chemical indicators that were inconsistent with the Moon forming 100 million years or later after the formation of the Solar System, the really clear indication came from the uranium-lead dating of zircon fragments recovered from the Apollo 14 mission. According to the study, the Moon was formed no later than 60 million years after the Solar System was born, with a minimum estimate of around 50 million years. To be precise, the study pegged 4.51 billion years ago as the time when the crust of the Moon was differentiated, or cooled down over the magma ocean.

Scientists from the University of Cologne recently published dates of the formation of the moon, which dated to around 4.51 billion years ago, matching the minimum estimate provided by the SSERVI researchers. The team from Cologne used samples that were collected over multiple missions. The scientists studied the relationship between uranium, hafnium and tungsten, all rare elements. These investigations allowed them to understand the formation of the mare basalt areas on the surface of the Moon, or the dark patches. Hafnium and tungsten act as radioactive clocks, with hafnium-182 decaying into tungsten-182. This decay took place only till about 70 million years after the formation of the Solar System. The abundance of tungsten-182 in samples is an indication of the timing of core formation, as it is formed by the melting of the silicate mantle, or the magma ocean. Dr Raul Fonseca from the University of Cologne explains, "By comparing the relative amounts of different elements in rocks that formed at different times, it is possible to learn how each sample is related to the lunar interior and the solidification of the magma ocean". A similar technique has been used on Martian asteroids, to date the formation of a crust of Mars as well.

The measurement of the tungsten and hafnium samples in the samples returned from the Apollo missions, allowed the Cologne researchers to date allow scientists to glimpse into the very origins of the Solar System, since such studies are not possible if entirely based on the rocks from Earth.

Most of the answers we have about the age of the Moon are obtained from the lunar samples returned from the Apollo missions. A total of 380 kg of samples were collected over the course of the Apollo missions, with over 90 kg being brought back on the Apollo 17 mission alone. Even after fifty years, scientists continue to study these samples, which form the basis of new or ongoing studies. Often, multiple studies are based on the same samples, which are referenced by numbers. Some of the rocks are locked away, and have never interacted with the atmos-



The surface of the Moon, soon after its formation

the formation of the Moon to about 50 million years after the formation of the Solar System. Professor Dr Carsten Munker, senior author of the study stated, "This age information means that any giant impact had to occur before that time, which answers a fiercely-debated question amongst the scientific community regarding when the Moon formed".

Determining the age of the Moon allows us to better understand the conditions in the Early Solar System. Such studies better help understand the formation and evolution of the Earth, as well as the series of events that led to the origin of life on the planet. While there are many geological processes that continue to reshape the surface of the Earth, the Moon is geologically inactive. The material on the Moon

phere of the Earth at all. In the 70s, the technological and scientific advances of the future were anticipated well in advance, and these untouched samples may provide new insights when processed by new and improved instruments and techniques.

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